Huffman codes

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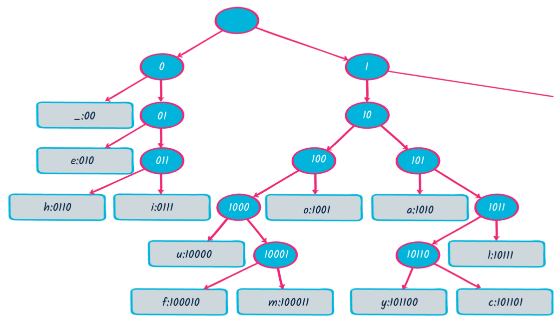
XPERT

Problem: How can we take advantage of unbalanced statistics to compress information?

When symbols are repeated (as in digital communications) and frequencies are available, Shannon's theory of information tells us that **maximal compression on average** can be achieved through variable-length codes. For instance, if you want to represent English texts in a compact binary code, you should choose a shorter code for 'E' than for 'Z'. Huffman codes do this optimally.



You can generate a Huffman code online.

Huffman codes

we obtained the following code

for letters in English.

You can check easily that the letters

with the shortest code are the ones you

expected to be the most frequent!

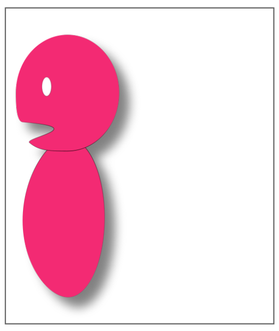
Conclusion:

Huffman code is a frequency-based

coding method which guarantees

the shortest code length on£average.

Because complexity is a minimum over all computations. And it may change through time.



Previous

Next